**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Procedures and Variable Scope

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What distinguishes a function from a procedure?

2. What is SMTP used for?

A) Downloading web pages
B) Sending email
C) Transferring file archives
D) Looking up domain names

3. "Cloud storage guarantees that files remain accessible during every Internet outage." Correct this. Explain why.

4. A school buys network devices from different makers. "Standards mean all devices must have the same maker." Correct this. How do standards help?

## Today's objectives

Use procedures, parameters and local and global variables. Use the technique in a different problem.

## Key vocabulary - Read, cover, recall

**Procedure**
A named subprogram that performs a task without returning a function value.

**Parameter**
A named input to a procedure or function.

**Local variable**
A variable created inside a subprogram, ordinarily limited to its scope.

**Global variable**
A variable declared for use across program sections, which should be handled carefully.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Procedure** | A named subprogram that performs a task without returning a function value. In the welcome message, identify where this technique applies. |
| **Parameter** | A named input to a procedure or function. In the ticket confirmation, identify where this technique applies. |
| **Local variable** | A variable created inside a subprogram, ordinarily limited to its scope. In the club announcement, identify where this technique applies. |
| **Global variable** | A variable declared for use across program sections, which should be handled carefully. In the text output routine, identify where this technique applies. |
| **Local and global constants as well as variables** | Local variables/constants belong to local scope; global ones have wider scope. Too much mutable global state makes bugs harder to trace. |

## I Do - One worked example

**A welcome message uses procedure and parameter. Explain what each contributes. [4 marks]**

1. Identify the program's purpose and the pieces of data needed.

2. Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

3. Parameter [1]; A named input to a procedure or function. [1].

4. Each feature needs a correct technical fact plus an explanation.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A ticket confirmation needs procedure. Explain the purpose. [2 marks]

2. Explain parameter for a club announcement. [2 marks]

3. When programming a text output routine, explain why local variable and global variable matter. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Write the call that displays a greeting for "Mia".**

```
procedure greet(person)
print("Hello " + person)
endprocedure
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain parameter and local variable in a sports registration. [4 marks]

Q4. A programmer makes a library reminder. Explain the roles of procedure, local variable and global variable. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A procedure is a named subprogram that performs a task without returning a value. ☐ T ☐ F
2. A named subprogram that performs a task without returning a function value ☐ T ☐ F
3. A named input to a procedure or function ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** A function returns a value to the caller. A procedure performs a task without returning a function value. ☐

**Q2** B - Sending email ☐

**Q3** Storage, software or processing on remote servers, accessed over a network. Access depends on the connection and service. ☐

**Q4** Agreed standards let hardware and software from different makers work together and exchange data. ☐

## Code First - Model

```
procedure greet(person)
print("Hello " + person)
endprocedure
greet("Mia")
```

**Output: Hello Mia**

## We Do - Check the model

1. Procedure [1]; A named subprogram that performs a task without returning a function value. [1]. ☐

2. Parameter [1]; A named input to a procedure or function. [1]. ☐

3. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Procedure [1]; Parameter [1]. ☐

**Q3** Parameter [1]; A named input to a procedure or function. [1]. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. ☐

**Q4** Procedure [1]; A named subprogram that performs a task without returning a function value. [1]. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**